

Machine Learning Basics

A concise introduction to supervised and unsupervised learning

1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence in which algorithms learn patterns from data and use those patterns to make predictions, classifications, or other decisions.

2. Supervised Learning

In supervised learning, a model is trained using examples that contain input features and known target values. Common tasks include classification and regression. Examples include spam detection, house-price prediction, and fraud detection.

3. Unsupervised Learning

Unsupervised learning works with data without predefined target labels. The goal can be to discover groups, structures, or representations in the data. Clustering and dimensionality reduction are common examples.

4. Model Training

A typical workflow includes data collection, cleaning, exploratory analysis, feature preparation, train/test splitting, model training, evaluation, and deployment. Keeping evaluation data separate helps estimate how the model behaves on unseen examples.

5. Evaluation Metrics

Classification may use accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Regression commonly uses MAE, MSE, RMSE, and R^2 . The appropriate metric depends on the problem and the cost of different types of errors.

6. Practical Considerations

Real-world machine learning requires attention to missing values, class imbalance, data leakage, feature quality, overfitting, reproducibility, and monitoring after deployment.

This document is provided as downloadable project/reference material.